

MASTER ÉCONOMIE APPLIQUÉE — IDEEP

Université de Lille

Can Canada Become a Strategic Alternative for Europe's Critical Minerals Supply?

Analysis of Canada's potential for six strategic minerals: nickel, lithium, copper, cobalt, graphite and rare earth elements

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Personal study — applied economics, industrial strategy and natural resources

July 2026

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1. Introduction

Since the early 2020s, critical minerals have occupied a growing place in economic and geopolitical debates. Nickel, lithium, copper, cobalt, graphite and rare earth elements have become indispensable components of the energy transition: they are used in the manufacture of batteries, electric vehicles, wind turbines, electricity grids and many so-called clean technologies. According to the International Energy Agency, global demand for these minerals accelerated markedly in 2024, rising by about 30% for lithium and 6 to 8% for nickel, cobalt, graphite and rare earths, growth directly linked to the rise of electric vehicles, energy storage, renewables and grids.

This growing importance is accompanied by rising awareness of the risks weighing on these supply chains. Production, and even more so processing, of these minerals remains concentrated in a small number of countries. The International Energy Agency notes that refining concentration remains very high and that China holds a dominant position there, which exposes importing economies to risks of export restrictions or supply disruption (IEA, 2025). Faced with this observation, the European Union is actively seeking to diversify its sources of critical mineral supply, rather than depending on a limited number of suppliers.

Canada has positioned itself as a potential partner in this diversification strategy. The Canadian government has published a national strategy on critical minerals, stating its ambition to develop responsible value chains “from exploration to recycling” and identifying six priority minerals for its international partnerships: nickel, lithium, cobalt, copper, graphite and rare earth elements (Government of Canada, 2022). These are precisely the minerals of interest to Europe for its own industrial needs.

This observation raises a question that seems simple, but whose answer is in fact more complex than it appears: can Canada become a strategic alternative for Europe's critical mineral supply? It would be tempting to answer affirmatively based solely on the fact that Canada holds significant geological resources. This reasoning, however, is misleading. As the OECD review on supply chain resilience recalls, the availability of a resource is not enough to guarantee reliable supply: the resource must also be extracted, processed, refined, and integrated into operational industrial chains (OECD, 2025).

The hypothesis defended in this report is therefore as follows: Canada holds real strategic potential for Europe, but this potential varies considerably depending on the mineral considered. It appears particularly credible for nickel and lithium, more partial for copper and cobalt, and still immature for graphite and rare earth elements. The main issue is therefore not merely the presence of resources, but Canada's ability to build complete value chains, from extraction to recycling, through processing and refining.

To examine this hypothesis, this report draws on public data from Natural Resources Canada, on Canada's critical minerals strategy, as well as on work by the OECD and the International Energy Agency devoted respectively to supply chain resilience and global critical minerals outlooks. It first proposes a conceptual framework for understanding the concepts used, before detailing the analytical methodology adopted, examining each of the six minerals, and then presenting the results of a

comparative scoring exercise built from this data. The report concludes with a discussion of Canada's place in a European diversification strategy, followed by a series of recommendations.

2. Conceptual framework: critical minerals, value chains and resilience

2.1. What are critical minerals?

The term “critical mineral” refers to mineral substances considered essential to the functioning of an economy, while being exposed to a significant risk of supply disruption or restriction. The criticality of a mineral therefore does not depend solely on its geological scarcity. A relatively abundant mineral in the Earth's crust can be classified as critical if its production or processing remains concentrated in a very small number of countries, or if its use is difficult to substitute in sectors considered strategic, such as energy, electronics, transportation or defence.

This definition explains why nickel, lithium, cobalt, copper, graphite and rare earth elements are among the priority minerals identified by Canada's critical minerals strategy (Government of Canada, 2022). These six substances share a common feature: they are indispensable to energy transition technologies, whether lithium-ion batteries, permanent magnets used in wind turbines and electric motors, or electricity distribution networks. The Government of Canada indeed presents its strategy as an effort to cover the entire industrial cycle, “from exploration to recycling,” signalling that the challenge is not limited to mining extraction.

2.2. Why the value chain matters as much as the resource

The value chain of a critical mineral comprises several successive stages: geological exploration, mining extraction, processing of raw ore into concentrate, refining to obtain an industrially usable product, manufacturing components or materials from this refined product, and finally, at the end of the cycle, recycling of end-of-life materials. Each of these stages requires specific skills, infrastructure and investment, and mastery of one stage in no way guarantees mastery of the following stages.

This is precisely the observation that structures the analysis proposed in this report. A country may hold considerable geological reserves without being able to industrially valorize them, for lack of processing or refining capacity. Conversely, a country can play a decisive role in refining a mineral without being its main mining producer, as is the case for China for several minerals examined in this report. The International Energy Agency's Global Critical Minerals Outlook 2025 insists on this point: refining concentration constitutes a risk distinct from extraction concentration, and often more difficult to correct in the short term, because refining capacity requires long and costly investments (IEA, 2025).

Assessing Canada's strategic potential therefore requires systematically distinguishing three notions: geological potential, i.e. the existence of resources and reserves; current production, which measures actual extraction capacity; and industrial capacity, which covers processing, refining and recycling infrastructure. This distinction, which structures the entire report, helps avoid a hasty conclusion of the type “Canada has resources, so it can replace current suppliers,” which would not withstand examination of the data.

2.3. Resilience, diversification and supply concentration

The OECD review on supply chain resilience proposes a useful framework for understanding the challenges facing Europe (OECD, 2025). Resilience is defined there as a supply chain's ability to absorb shocks, whether geopolitical, logistical or economic in origin, without lasting interruption of access to essential goods. Yet the concentration of imports among a limited number of suppliers is one of the main fragility factors identified by the OECD: the more an economy depends on a small number of sources for a strategic good, the more vulnerable it is to a unilateral decision, a local disruption, or a conflict affecting that supplier.

Faced with this risk, the OECD recommends a diversification strategy rather than a complete relocation of production chains (OECD, 2025). Diversification does not mean entirely replacing a dominant supplier with another, but rather multiplying supply sources in order to reduce dependence on a single origin and increase room for manoeuvre in the event of a shock. This approach is economically realistic, since a full relocation of critical mineral value chains would be extremely costly and lengthy to implement, given the investments required in refining and processing.

It is within this framework that Canada can be considered as a partner of the European Union. The question is not whether Canada can fully substitute for Indonesia in nickel, the Democratic Republic of Congo in cobalt, or China in graphite and rare earths. Rather, it is a matter of assessing, mineral by mineral, whether Canada can constitute a credible additional source, capable of reducing the current concentration of European supply. This is the analytical framework this report applies through the analysis of the six minerals selected.

3. Data and methodology

This report draws on four main institutional sources. Factual data by mineral come from the “Minerals and metals facts” sheets published by Natural Resources Canada (Natural Resources Canada, 2026), which detail for each substance the uses, Canadian and global production, reserves, foreign trade, recycling and international context. Canada's critical minerals strategy, published by the Government of Canada (Government of Canada, 2022), helps understand the country's industrial objectives and the rationale used to prioritize certain minerals. The OECD review on supply chain resilience provides the conceptual framework relating to diversification and risk concentration (OECD, 2025). Finally, the International Energy Agency's Global Critical Minerals Outlook 2025 provides a global perspective on demand, refining concentration and geopolitical issues (IEA, 2025).

The analysis covers six minerals selected because they are among the priorities of the Canadian strategy and also correspond to European needs in terms of energy transition: nickel, lithium, cobalt, copper, graphite and rare earth elements. For each, the report examines strategic uses, the Canadian situation, the global production context, as well as Canada's specific strengths and limitations.

In order to structurally compare the relative potential of these six minerals, a qualitative scoring exercise was built from the institutional data presented above. It is important to clarify from the outset the nature of this exercise: the scores assigned do not constitute raw official data and do not claim to precisely measure a quantified economic reality. This is a qualitative assessment, built through

reasoned interpretation of available information, whose objective is to compare the relative strategic potential of the minerals with one another, not to produce an absolute measure of their economic value.

Six criteria were selected for this scoring, each rated on a scale of one to five, where 1 corresponds to a very low level, 2 to a low level, 3 to a medium level, 4 to a strong level and 5 to a very strong level. Available resources measure the scale of deposits and reserves identified in Canada. Current production reflects the volume actually extracted and Canada's rank in global production. Projects under development account for the number and maturity of ongoing mining projects, whether at the exploration, economic assessment or construction stage. Transformation capacity assesses the existence of refining, conversion or processing infrastructure on Canadian territory. Alignment with European needs measures the relevance of the mineral to the industrial sectors Europe seeks to secure, particularly batteries and clean technologies. Strategic reliability, finally, reflects Canada's institutional and political stability as a trading partner.

These six criteria do not carry equal weight in the final score. Available resources, current production and transformation capacity each count for 20% of the score, as they determine Canada's real and immediate capacity to contribute to supply. Projects under development count for 15%, as they indicate a future trajectory rather than a present capacity. Alignment with European needs also counts for 15%, this criterion linking Canadian potential to the European Union's strategic demand. Strategic reliability, finally, counts for only 10%, not because it is secondary, but because it varies little from one mineral to another: Canada remains, regardless of the mineral considered, a politically stable country and an ally of Europe, which limits this criterion's ability to differentiate between the six substances.

The final score for each mineral is obtained through a weighted average of the six criteria, brought to a base of one hundred points according to the following formula: $\text{Score} = ((\text{Available resources} \times 20) + (\text{Current production} \times 20) + (\text{Projects under development} \times 15) + (\text{Transformation capacity} \times 20) + (\text{Alignment with European needs} \times 15) + (\text{Strategic reliability} \times 10)) / 5$. Four levels were defined to interpret this score: a very strong strategic potential above 80 points, a strong but incomplete potential from 65 points, a medium potential from 50 points, and a limited short-term potential below this threshold.

This method has limitations that should be explicitly acknowledged. The scoring remains qualitative and relies on the interpretation of public data, without access to detailed financial data such as CAPEX or OPEX, nor a company-by-company analysis. Nor does it constitute a complete prospective modelling of production trajectories. The scores obtained should therefore be read as comparative indicators, useful for ranking the six minerals relative to one another, and not as absolute measures of profitability or industrial feasibility.

The table below summarizes the scores assigned to each mineral on the six criteria selected, as well as the final score and the resulting potential level. For ease of reading, this table is presented on the following page, in landscape format. These results are discussed in detail in sections 5 and 6.

Table 1 — Scoring of Canadian strategic potential by mineral

Scores assigned on a scale of 1 (very low) to 5 (very strong) for each of the six criteria selected; the final score is a weighted average brought to a base of 100 points (see methodology, section 3).

Mineral	Available resources	Current production	Projects under development	Transformation capacity	Alignment with EU needs	Strategic reliability	Score
Nickel	4	5	4	4	5	5	8
Lithium	5	3	4	3	5	5	8
Copper	3	4	3	3	5	5	7
Cobalt	3	2	3	4	5	5	7
Rare earth elements	5	1	4	2	5	5	6
Graphite	3	2	4	2	5	5	6

4. Analysis by mineral

This section examines each of the six selected minerals in turn. For each, the analysis presents the strategic uses of the substance, the Canadian situation in terms of production and reserves, the global production context, before identifying Canada's specific strengths and limitations and drawing an interpretation for the European diversification strategy.

4.1. Nickel

Nickel is a metal with multiple uses. It is used in stainless steel, various metal alloys, electroplating, and increasingly in the lithium-ion batteries used by electric vehicles. This diversity of uses distinguishes nickel from the other minerals examined in this report: in 2024, batteries still accounted for only about 15% of global nickel consumption, with most demand still linked to stainless steel production.

According to Natural Resources Canada data (2026), in 2024 Canadian mines produced 125,364 tonnes of nickel, placing Canada fourth among global mining producers. This production is spread across four provinces: Ontario, accounting for about 40%, Quebec, with about 37%, Newfoundland and Labrador, with about 14%, and Manitoba, with about 9%. This geographic distribution reflects the long-standing and diverse nature of Canada's nickel mining sector, established for several decades in the Canadian Shield.

Globally, however, production remains dominated by Indonesia, which alone accounts for about 61% of global nickel mining production, compared to 3.5 to 4% for Canada. This scale gap is considerable and should be kept in mind: Canada does not compete with Indonesia in volume. Its specificity lies elsewhere, in the maturity of its value chain. The country indeed has operational nickel refineries in Fort Saskatchewan, Sudbury and Long Harbour, allowing it to process part of its production on its own territory rather than exporting only raw ore. In 2024, Canadian exports of nickel and nickel-based products reached \$4.5 billion.

All these elements make nickel the most mature mineral in Canada's portfolio. Canada combines significant production, long-standing industrial know-how, established refining capacity and recognized geopolitical stability. It is not a supplier capable of substituting for Indonesia on a global scale, but a credible and immediately operational diversification source for Europe, particularly for uses related to batteries and stainless steel.

4.2. Lithium

Lithium is one of the most emblematic minerals of the energy transition, owing to its central role in rechargeable batteries. In 2024, batteries accounted for about 87% of global lithium demand, a proportion that illustrates how inseparable this mineral has become from the rise of electric vehicles and energy storage.

The Canadian situation presents a mixed profile. According to Natural Resources Canada (2026), in 2024 the country had only two active lithium mines, located in Manitoba and Quebec, for total production of about 5,983 tonnes. This volume places Canada seventh in the world, with only 2.5% of

production, far behind Australia, the top producer with 37.1% of the world total, Chile, with 20.7%, or China, with 17.3%. This modest current production nonetheless contrasts sharply with the scale of resources identified on Canadian territory.

Canada indeed has hard-rock deposits estimated at about 5.4 million tonnes of lithium oxide, in addition to brine resources located in Alberta and Saskatchewan, estimated at about 8.6 million additional tonnes. In total, Canadian lithium resources amount to approximately 14 million tonnes of lithium oxide, or about 6.5 million tonnes of contained lithium, and the country holds about 4.4% of world reserves. Several projects are under development, notably around Val-d'Or in Quebec, where the company North American Lithium has resumed production, and the Nemaska Lithium project, which plans to exploit the Whabouchi mine as well as build a lithium hydroxide refinery in Bécancour.

This last point is essential for correctly interpreting Canadian potential: the value of a lithium project is not measured solely by the size of the deposit, but by the ability to convert the extracted ore into lithium hydroxide or carbonate, forms directly usable by the battery industry. Yet Canada remains today a net importer for certain lithium-based products, and its recycling, although developing, remains limited. Canadian lithium potential is therefore real, but still largely under construction: resources and projects exist, while production and industrial processing still need to be consolidated. For Europe, this means Canadian lithium constitutes a promising medium-term strategic bet, more than an immediately available large-scale supply source.

4.3. Copper

Copper occupies a special place among critical minerals: it is not a new or emerging metal, but a long-established industrial metal whose strategic importance has been renewed with the electrification of the economy. Copper is indispensable for electricity grids, construction, industrial equipment, electric vehicles and solar energy. It also has the advantage of being recyclable without major loss of quality, making it a unique case among the minerals studied in this report.

According to Natural Resources Canada (2026), in 2024 Canadian mines produced 514,582 tonnes of copper concentrate, a volume down 26.2% compared to 2015, when production reached 697,322 tonnes. This declining trend deserves attention, as it contrasts with the dynamic observed for nickel or lithium. British Columbia remains the top producing province, with about 48% of national production, well ahead of Quebec and Newfoundland and Labrador.

Globally, Canada ranks twelfth among copper producers, with about 2% of global mining production, far behind Chile, the top producer with 5.3 million tonnes, or 23% of the roughly 23 million-tonne world total. Canadian reserves, estimated at 8.3 million tonnes, represent only about 1% of world reserves. Canada nonetheless remains a significant exporter: in 2024, Canadian exports of copper and copper-based products reached \$10.7 billion, mainly to the United States, for total trade of \$17.3 billion.

Canadian copper thus constitutes a solid value chain useful for electrification, but of modest weight on a global scale. Unlike nickel, it cannot be presented as an already mature and internationally competitive value chain: the production decline observed since 2015 even calls for caution. For Europe, Canadian copper represents a credible complementary supplier, capable of fitting into long-term

supply partnerships, without however constituting a dominant source or a leading alternative to the major South American producers.

4.4. Cobalt

Cobalt is used in batteries, in certain high-temperature superalloys, in cutting tools, in magnets and in several industrial applications, including pigments. Its particularity, in Canada as elsewhere, is that it is most often produced not as a primary mineral, but as a by-product of extracting other metals, in this case nickel.

According to Natural Resources Canada (2026), in 2024 Canadian mines produced about 3,351 tonnes of cobalt, distributed among Quebec, with 35% of the total, Ontario, with 33%, Newfoundland and Labrador, with 28%, and Manitoba, with 4%. This distribution largely overlaps with the geography of Canadian nickel, confirming the close link between these two value chains. Canada also produced about 5,920 tonnes of refined cobalt, thanks to three refineries located in Fort Saskatchewan, Port Colborne and Long Harbour, reflecting real, though modest-scale, processing capacity.

Canada ranks seventh in the world among cobalt producers, with about 1.2% of global mining production. The Democratic Republic of Congo dominates this market by far, with 220,000 tonnes in 2024, or 76.6% of the world total, followed at a distance by Indonesia, with 9.8%. World reserves follow a similar distribution, with the DRC accounting for about 52.8%, compared to about 1.9% for Canada. In 2024, Canadian exports of cobalt and cobalt-containing products reached \$344 million, a much smaller amount than for nickel or copper.

Canadian cobalt is therefore useful, but structurally limited. Canada cannot claim to compete with the Democratic Republic of Congo in extraction, nor with China in refining, the latter dominating global processing of cobalt destined for batteries. The appeal of Canadian cobalt for Europe lies less in volumes than in its institutional reliability, its close link with the already mature nickel value chain, and the existence of local refining capacity. It represents a complementary diversification source, rather than a leading alternative to the main global producers.

4.5. Graphite

Graphite is used in industrial electrodes, refractory materials, lubricants and foundry products, but its most strategic use today concerns lithium-ion battery anodes, where it is actually the heaviest component by weight of a conventional battery. The growth of electric vehicles therefore directly drives global graphite demand, just as it does for lithium or nickel.

The Canadian situation in this area nonetheless remains modest. In 2024, only one graphite mine was operating in Canada, the Lac des Îles mine in Quebec, producing 11,697 tonnes of natural graphite, or about 12,000 tonnes according to Natural Resources Canada's rounded data. This volume places Canada eighth in the world, with only 0.7% of global natural graphite production, estimated at 1.6 million tonnes in 2024. China dominates this market by far, with about 1.27 million tonnes produced, or 79.4% of the world total.

Canadian graphite reserves, estimated at 5.9 million tonnes out of a world total of 290 million tonnes, place the country tenth in the world, far behind China, Brazil and Madagascar. In 2024, Canada exported \$25.8 million worth of natural graphite and \$18.1 million worth of synthetic graphite, mainly to the United States. Graphite recycling, particularly from end-of-life lithium-ion batteries, remains underdeveloped.

Graphite thus clearly illustrates the gap that can exist between a mineral's strategic importance and the maturity of the value chain that produces it. Canada has a mining base, with the Lac des Îles deposit, but its production and processing capacity lag well behind Chinese dominance. The decisive challenge for this value chain's future lies not in raw extraction, but in scaling up higher-value processed materials, such as coated spherical graphite used directly in battery anode manufacturing.

4.6. Rare earth elements

Rare earth elements comprise seventeen chemical elements, including the lanthanides, as well as scandium and yttrium. They are used in the manufacture of permanent magnets, catalysts, polishing powders, ceramics, specialty glass, lasers and numerous high-technology applications. Permanent magnets are today the main use for these elements, accounting for about 47.6% of their use in 2024, a central application for electric vehicles, wind turbines and electronics.

Canada presents, for these elements, a singular profile among the six minerals studied: the country has considerable known resources, estimated at over 15.2 million tonnes of rare earth oxides, but is not yet a significant commercial producer. Several projects are under development across the country, at various stages ranging from exploration to preliminary economic assessment and feasibility studies, but none has yet reached the stage of large-scale industrial production.

This situation contrasts sharply with China's position, which produced about 270,000 tonnes of rare earth elements in 2024, or 69% of global mining production, and which also concentrates about 90% of global refined production. The United States, Myanmar, Australia and Thailand round out the ranking of producers, but at levels far below China's. This double dominance, in both extraction and refining, is partly explained by the complexity of the separation processes required to isolate each of the seventeen elements, industrial know-how that China has developed and consolidated over several decades.

For rare earths, Canada's main obstacle is therefore not the geological resource, which is abundant, but separation and refining capacity, which still needs to be built. Canadian potential thus appears strategic in the long term, particularly if several projects currently under development reach the production stage, but it would be premature to consider Canada as an operational alternative to China in the near term. For Europe, this represents a partner to support upstream, notably through investment in processing capacity, rather than an immediately available supplier.

5. Scoring results and strategic ranking

Applying the methodology presented in section 3 to the six minerals studied results in the following ranking, summarized in the table below.

Rank	Mineral	Score /100	Level
1	Nickel	89	Very strong strategic potential
2	Lithium	81	Very strong strategic potential
3	Copper	74	Strong but incomplete potential
4	Cobalt	70	Strong but incomplete potential
5	Rare earth elements	69	Strong but incomplete potential
6	Graphite	65	Strong but incomplete potential

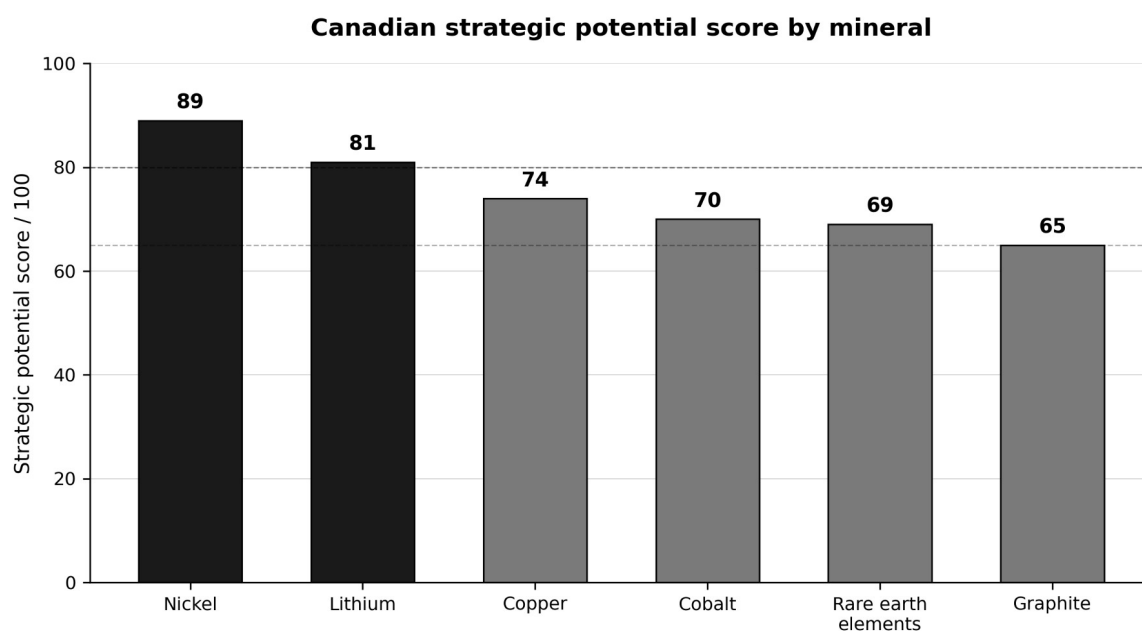


Chart 1 — Canadian strategic potential score by mineral

This ranking highlights two minerals clearly above the others: nickel, with a score of 89, and lithium, with a score of 81, both classified as having very strong strategic potential. The other four minerals fall within a narrower range, between 65 and 74 points, corresponding to the “strong but incomplete potential” category: copper, with 74 points, cobalt, with 70, rare earth elements, with 69, and graphite, with 65.

The gap between the first and second groups does not reflect a difference in the strategic interest of these minerals for Europe: all six substances are considered highly relevant to the energy transition, which is why they all receive a maximum score on the alignment-with-European-needs criterion. The difference lies elsewhere, in the current production and transformation capacity criteria, where nickel and, to a lesser extent, lithium, clearly outperform the other four minerals.

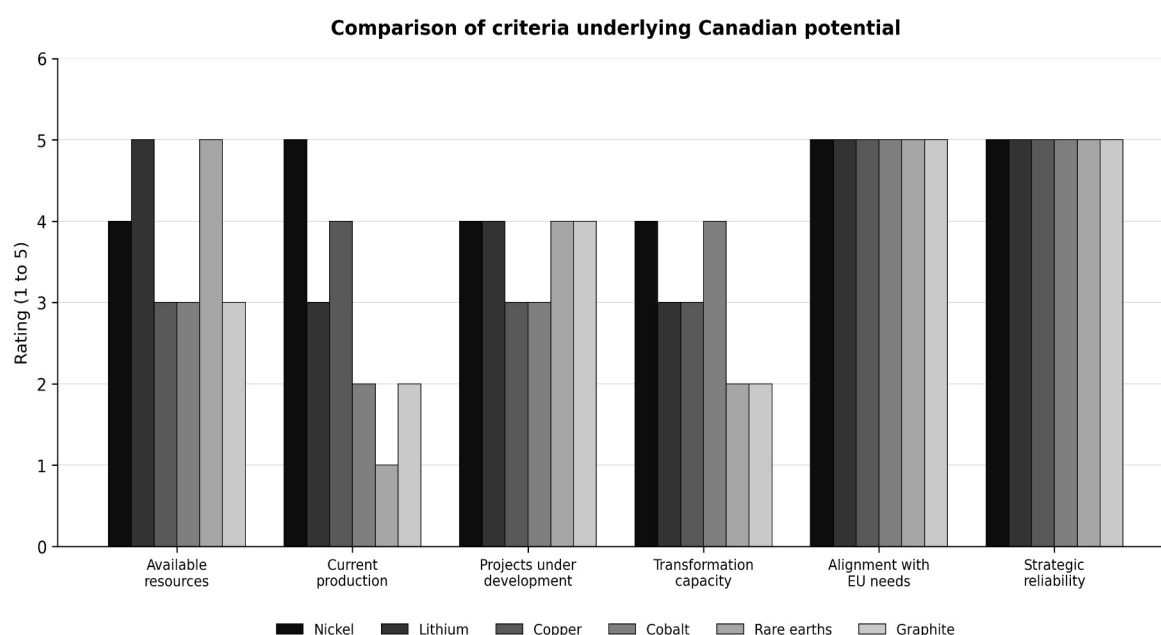


Chart 2 — Comparison of criteria underlying Canadian potential

The comparison of criteria by mineral, presented in chart 2, helps refine this reading. Nickel stands out with a balanced profile, with high and consistent scores across all criteria, reflecting a value chain that is both resource-rich, productive and already processed on Canadian territory. Lithium presents a different profile: very strong in available resources and projects under development, but more modest in current production and processing capacity, reflecting a value chain still under construction. Rare earth elements show the sharpest contrast in the portfolio, with a maximum score in available resources but a minimum score in current production, illustrating how geological abundance has not yet translated into industrial capacity. Graphite, finally, remains weak in both production and processing, despite a relatively high number of projects under development.

6. Industrial maturity and value chains

The central idea of this section can be summarized simply: a mineral's strategic potential does not depend solely on available resources, but also, and perhaps above all, on the industrial maturity of the value chain that valorizes it. To objectify this idea, an industrial maturity indicator was calculated for each mineral, as the average of the scores assigned to current production and transformation capacity.

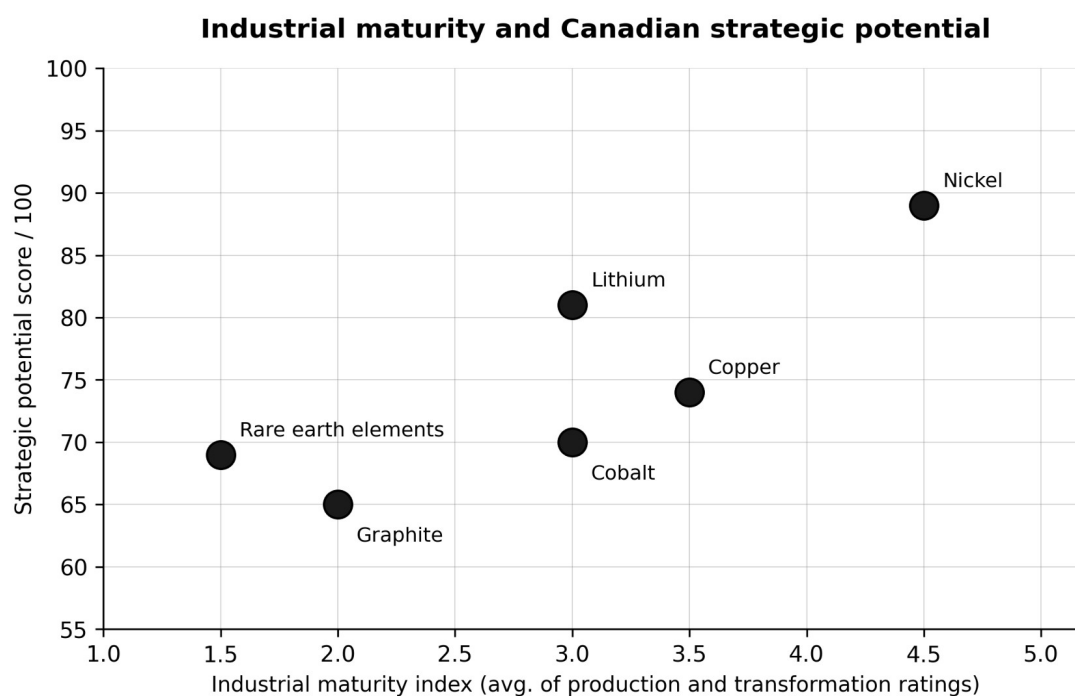


Chart 3 — Industrial maturity and Canadian strategic potential

This chart highlights differentiated trajectories depending on the mineral. Nickel combines a high strategic score with equally high industrial maturity, reflecting a value chain that is both recognized and already operational. Lithium stands out with a high strategic score combined with more moderate industrial maturity, confirming the diagnosis established in section 4: the potential is real, but industrial processing still needs to be consolidated. Copper occupies an intermediate position, consistent with a solid value chain of modest global weight. Cobalt benefits from adequate maturity thanks to its refining capacity, despite limited production. Rare earth elements and graphite, finally, appear as the two least industrially mature minerals, even though their strategic importance for Europe remains high, making them the most emblematic cases of the gap between geological potential and industrial capacity.

This gap is not unique to Canada: it corresponds to a broader phenomenon, documented by the International Energy Agency, according to which the concentration of global refining today constitutes a risk at least as significant as the concentration of mining extraction. In Canada's case, this observation calls for nuance in any purely geological reading of the country's strategic potential: having deposits is not enough to guarantee a significant contribution to European supply, as long as the corresponding processing capacity is not developed.

7. Discussion: Canada as a diversification partner for Europe

All the elements presented in the preceding sections make it possible to answer the initial research question in a nuanced way. Canada can become a partial and credible strategic alternative for Europe's critical mineral supply, but it cannot constitute a complete substitute for currently dominant suppliers. This conclusion plays out differently depending on the mineral, and it is useful to distinguish three groups.

The first group brings together the priority and immediately credible minerals: nickel and lithium. Canadian nickel already has a mature value chain, combining production, refining and institutional stability, making it the strongest candidate for enhanced cooperation with Europe in the short term. Lithium, although less advanced in terms of current production, presents resources and projects significant enough to justify priority investment in its industrial processing.

The second group corresponds to useful but partial diversification: copper and cobalt. These two minerals offer Canada a role as a complementary supplier, capable of contributing modestly but reliably to European diversification, without however rivalling the major global producers, namely Chile for copper or the Democratic Republic of Congo for cobalt.

The third group brings together minerals whose strategic potential still needs to be built: graphite and rare earth elements. For these two substances, the abundance of Canadian resources has not yet translated into significant industrial capacity, particularly in the processing and refining segments, where China retains a considerable lead.

This differentiated reading directly aligns with the OECD's recommendations on supply chain resilience (OECD, 2025). The organization notes that diversification is preferable to a full relocation of production chains, an option deemed costly and unrealistic on a large scale. From this perspective, Canada should not be analyzed as a replacement for current suppliers, but as one link in a deliberately diversified supply chain, alongside other partners. This approach implies accepting residual dependence on certain dominant producers, while reducing the overall vulnerability of the system through the multiplication of sources.

The International Energy Agency's Global Critical Minerals Outlook 2025 offers a complementary perspective on this discussion, noting that supply risk is not located solely at the extraction level, but also, and sometimes especially, in the intermediate refining and processing segments (IEA, 2025). This observation reinforces the report's central argument: Canada's contribution to European supply security will depend less on the scale of its geological resources than on its ability to develop, mineral by mineral, complete industrial links, from extraction to recycling.

8. Recommendations

The following recommendations follow directly from the analysis conducted in this report and are presented mineral by mineral, followed by a cross-cutting recommendation on structuring partnerships.

Nickel

Since Canada's nickel value chain is already mature, with significant production and established refining capacity, it constitutes the natural pillar of enhanced cooperation between Europe and Canada, particularly for uses related to batteries and stainless steel.

Lithium

Canada's lithium potential justifies targeted support, not so much for extraction, already underway through several projects, but for industrial processing into lithium hydroxide or carbonate, a necessary condition for this geological potential to translate into industrial supply usable by European battery value chains.

Copper and cobalt

These two minerals can be mobilized as complementary diversification sources, within the framework of long-term supply partnerships, without Europe considering Canada a dominant supplier for either of these metals.

Graphite and rare earth elements

These two value chains require priority investment in separation, refining and processing segments, particularly toward coated spherical graphite intended for battery anodes. Without such an effort, the abundance of resources identified in Canada will remain largely theoretical from the standpoint of European supply.

Cross-cutting recommendation

More broadly, partnerships between Europe and Canada would benefit from being designed across the entire value chain, not solely around extraction: this requires jointly integrating processing, recycling, long-term supply contracts, industrial infrastructure financing and environmental standards, in order to build a stable and mutually beneficial relationship rather than a simple flow of raw material exports.

9. Conclusion

This report set out to examine whether Canada can become a strategic alternative for Europe's critical mineral supply. The analysis conducted on six priority minerals, nickel, lithium, copper, cobalt, graphite and rare earth elements, shows that the answer can be neither entirely positive nor entirely negative, but must be differentiated according to the mineral considered.

Canada already appears credible for nickel, whose value chain combines production, refining and institutional stability. It proves very promising for lithium, where resources and projects are numerous, but where industrial processing still needs to be consolidated. It constitutes a useful but partial partner for copper and cobalt, two solid value chains of modest weight on a global scale. It remains, finally, under construction for graphite and rare earth elements, two minerals whose geological potential is real, but whose industrial capacity, particularly in the separation and refining segments, largely remains to be developed.

The main lesson of this work is that mineral sovereignty cannot be reduced to geology. It depends, at every stage, on the ability to process, refine and secure complete value chains. It is this industrial capacity, more than the abundance of resources, that will determine, in the years to come, Canada's real place in Europe's strategy of diversifying its critical mineral supply.

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